

1. Given marks = [45, 67, 45, 89, 67, 90], print how many marks were entered in total and how many of them are unique.

*Hint: len() on the list gives one number, len() on set(marks) gives the other.*

2. Create an empty set, add the languages "Python", "SQL" and "Python" again, then print the number of languages stored. Explain in a comment why it is not three.

*Hint: set() makes an empty set — {} would give you a dictionary. Adding a duplicate is silently ignored.*

3. Given a sentence typed by the user, print how many unique letters it contains, ignoring spaces and treating upper and lower case as the same.

*Hint: lower() the sentence, replace the spaces with nothing, then wrap it in set().*

4. Given nums = {10, 20, 30, 40}, remove 20 safely, try to remove 99 without crashing the program, and print the remaining values in sorted order.

*Hint: discard() does nothing when the value is missing; remove() would raise a KeyError.*

5. Given a list of cities visited with several repeats, print the unique cities in alphabetical order, and then print the same list with duplicates removed but the original order kept.

*Hint: sorted(set(cities)) handles the first part. For the second, loop and append only when the city is not already in your new list.*